

How Texture-Inducing Contraction Vectors Affect the Fibrous Capsule Shrinkage Around Breasts Implants?

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Received: 17 August 2009 / Accepted: 11 February 2010 / Published online: 31 March 2010
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Abstract

Background Breast augmentation was performed on nine female volunteer patients to determine the influence of three textured surfaces on fibrous capsule formation. The high degree of surface texture reduced the capsular fibrous tissue formation.

Methods Patients were distributed according to the degree of texture created by the open-pore diameter and depth of the surface texture. Macroscopic and histological examinations were performed to evaluate the implant texture and fibrous capsule interface. Magnetic resonance imaging established a mutual correlation with breast firmness achieved by the Baker grade.

Results The parallelogram law was applied to linear vectors arising from the same initial point into the concavities or undulations on the textured surface to achieve the resultant contraction vector. They were identical to resultant vectors created on the fibrous capsule except in the inverted direction. The adhesive effect or mirror-image tissue response to the implant texture on the fibrous capsule noticeably reduced capsular contracture but only with macrotextured implants. Increased capsular contracture resulted from implants showing micro- and medium surface texture with no significant difference between them.

Conclusion Biocell™ implants created resultant vectors on the capsular interface of small and similar lengths with divergent directions creating natural breast firmness. Biodegradation of Polyurethane™ foam disrupted the

texture from creating resultant vectors of long and different lengths and in variable directions, increasing breast firmness from 12 months. Siltex™ implants created few and very long resultant vectors perpendicular to the fibrous capsule that developed fibrous capsule contraction after 9 months. We conclude that implants with macrotextured surfaces significantly reduced the risk of fibrous capsular contraction.

Keywords Capsular contracture · Implant roughness · Textured surface shell · Vectors

The development of a fibrous capsule around a breast implant is an inflammatory reaction of the body against a foreign material. Gradual contraction of the fibrous capsule distorts the implant in various degrees from mild breast firmness to a hardened sphere. Thickening and contraction of the fibrous capsule is a multifactor process involving the implant as well as clinical and surgical conditions. The surface of the shell and the gel cohesiveness are factors related to the implant [1]. Site of placement, bacterial infection, hematoma, and patient individuality are related to surgical and clinical conditions [2, 3]. Several studies have emphasized the influence of the implant's surface in preventing capsular contracture. Retrospective controlled trials concerning smooth and textured surfaces have suggested that implant texturization reduces the incidence of fibrous capsular contraction [4]. The open-pore diameter and depth of the surface texture noticeably influence fibrous capsule formation [5]. Linear vectors created on the surface texture induce capsular contraction or shrinkage. Vectors are described as directed straight lines with an origin and a length. They are graphically represented by a line with an arrowhead at its tip which indicates the length

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and direction of the vector [6]. Resultant vectors inducing capsular contracture are achieved by the sum of linear vectors created over the fibrous capsule.

The purpose of this study was to establish the influence of resultant vectors created by the open-pore diameter and depth of the implant surface texture on capsular contracture.

Patients and Method

Nine female volunteer patients, ranging in age from 28 to 48 years and with two pregnancies among them, underwent subglandular breast augmentation using an inframammary incision. Natrelle™ Biocell® Intrashiel™ 27-120301 (Allergan, Inc., Irvine, CA, USA), Siltex® RoundGel™ 354-4300 (Mentor Corp., Santa Barbara, CA, USA), and Polyurethane-Covered Foam™ 30621-305 (Silimed Polyurethane, Rio de Janeiro, RJ, Brazil) breast implants, round-shaped with high profile and cohesive silicone gel, were used. Implants had three different textures with respect to the depth and open-pore diameter on the surface texture. All patients signed an informed consent form agreeing to the treatment performed according to the Helsinki Declaration. Patients were evaluated every 3 months using the Baker grade. Magnetic resonance imaging examination was performed after 18 months. Open capsulectomy and replacement of the implants with Biocell Intrashiel™ 27-120301 implants was performed 18 months after the breast augmentation. Macro photos of the fibrous capsule were taken with a digital Sony camera using a 2.6× wide conversion high-grade lens VCL-DH1730. The fibrous capsule was divided into 5-mm² fragments, fixed with paraffin, and stained with hematoxylin and eosin for histological examination using an optical microscope with 100× and 400× Zeiss objectives.

Implant Examination

Strips 10 mm long and 1 mm wide were cut from the top and both sides of the circumference of the replaced implants. The fragments were used to identify open-pore diameter and the inclination and depth of the implant texture. They were not fixed or stained. Fragments were examined using a fluorescence optical microscope (Model Ellipse E600FM) with 10/20× objectives and 400× ocular augmentation. Photos were taken with a Nikon DX1200 digital camera and digitized using Nikon's Act-1 Platform version 2.7 software program.

Results

Three patients had 300-ml Biocell Intrashiel™ implants, three patients had 300-ml Siltex Round Gel implants, and

three patients had 305-ml Polyurethane-Covered Foam™ implants. Clinical evaluation was performed using the Baker grade for breast firmness and magnetic resonance imaging (MRI) to visualize the implant contour. Table 1 summarizes the Baker grade for mammary firmness on both breasts every 3 months for 18 months. The increase in breast firmness leading to Baker grade II began at month 9 only for the Siltex implants. Baker grade II was achieved for both Siltex™ and Polyurethane™ implants at month 12. Baker grade III occurred first with Siltex implants at month 15 and with both Siltex and Polyurethane implants at month 18. A soft and natural breast was achieved only with Biocell™ implants. MRI revealed a noticeable correlation with clinical breast firmness determined by the Baker grade. Baker grades I and III and the corresponding MRI examination are shown in Figs. 1 and 2, respectively.

Textured Surface Shell of the Implants

Siltex™ implants showed microtexture composed of undulations regularly distributed on the implant's surface (Fig. 3, left). Biocell™ implants showed macrotexture composed of concavities with a large open-pore diameter and depth that were irregularly arranged over the implant's surface (Fig. 3, right). Polyurethane™ exhibited irregular porosity of the texture which was composed of concavities with an enlarged open-pore diameter and depth, indicating disruption of the medium texture caused by biodegradation of the polyurethane foam (Fig. 4). Table 2 gives the morphology and texture of the implants.

Resultant Vectors on the Texture

On a Cartesian coordinate system, two linear vectors were drawn from the deepest point of the depression to the extremities of the open-pore diameter for macro- and medium textures corresponding to Biocell™ and Polyurethane™ implants, respectively. Resultant vectors were achieved using the parallelogram law for the addition of forces (Fig. 5, right, left). In Fig. 5 the parallelogram law was applied in a few concavities just to compare length and direction of the resultant vectors for Biocell (Fig. 5, right) and Polyurethane (Fig. 5, left) implants. For the micro-textured surface of Siltex™ implants, the horizontal axis of the Cartesian coordinate system was set tangentially to the base of the undulation. Several horizontal axes were drawn from the deepest to the more superficial undulations. The distance from the deepest axis to the others created perpendicular vectors in different points of the horizontal axis with coincident direction. The sum of the vectors created on the horizontal axis was projected over the vertical axis creating a resultant vector perpendicular to the implant surface (Fig. 6). In Fig. 4, the different colors indicate the

Table 1 Distribution of patients according to breast firmness every 3 months

	Siltex™												Polyurethane™											
	Right breast						Left breast						Right breast						Left breast					
	3	6	9	12	15	18	3	6	9	12	15	18	3	6	9	12	15	18	3	6	9	12	15	18
Baker I	3	3	2	1	0	0	3	3	2	2	0	0	3	3	3	2	1	0	3	3	3	2	1	0
Baker II	0	0	1	2	3	2	0	0	1	1	2	1	0	0	0	1	2	3	0	0	0	1	2	3
Baker III	0	0	0	0	0	1	0	0	0	0	1	2	0	0	0	0	0	0	0	0	0	0	0	0
Baker IV	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Polyurethane™												Biocell™											
	Left breast						Right breast						Left breast						Right breast					
	3	6	9	12	15	18	3	6	9	12	15	18	3	6	9	12	15	18	3	6	9	12	15	18
Baker I	3	3	3	2	0	0	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
Baker II	0	0	0	1	3	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
Baker III	0	0	0	0	0	2	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0
Baker IV	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

vector for each undulation, whose sum was the resultant vector. The magnitude of the resultant vectors is indicated by the length of the arrow and direction is indicated by the arrowhead, characterizing the vector displacement. Micro- and medium textures showed predominance of long and different resultant vectors with variable directions, inducing a large displacement for the fibrous capsule. Macro-texture created similar and small resultant vectors with divergent directions, avoiding the displacement of the fibrous capsule.

Fibrous Capsule Examination

No silicone gel particles were found in the fibrous capsules around the three different implant textures. Table 3 summarizes the macroscopic and histological findings for Biocell™ implants which imprinted a texture mirror image on the interface with the capsule (Figs. 7, 8). Capsular fibrous tissue for Polyurethane™ implants showed an inflammatory reaction around foam fragments causing an irregular and disorganized fibrous tissue distribution. The fibrous capsule was thicker for macro- and medium-textured implants than for microtextured implants. The resultant vectors arising from the fibrous capsule went into the concavities or undulations of the capsular interface. They had the same length but the opposite direction as the resultant vectors created on the implant texture.

Discussion

Resultant vectors acting on the fibrous capsule are a mirror image of resultant vectors produced by the implant texture except in the opposite direction. The contraction force is exerted according to the length of the resultant vector. It is represented by the displacement from an initial point at the fibrous capsule to another point determined by the arrowhead, directed to the center of the implant. The contraction force of resultant vectors significantly decreased as the distance increased from the center of the implant [7]. Normal breast firmness achieved for Biocell™ implants correlated with the short length of the resultant vectors far enough from the implant center. Otherwise, resultant vectors of equal length and divergent directions tend to neutralize each other, thus avoiding capsular contracture [8]. This is in concordance with the inverse property of vector addition since the sum of resultant vectors of the same length and opposite directions is equal to zero. Small vectors appear structurally connected to the existence of a critical open-pore diameter of the texture of greater than 150 μm [9, 10]. Table 2 indicates an open-pore diameter of greater than 150 μm for Biocell™ and Polyurethane implants. However, the effectiveness of the polyurethane

Fig. 1 *Right* Baker grade I for both breasts with Biocell™ implants after 18 months. *Left* Magnetic resonance imaging showing regular contour for the implant surface

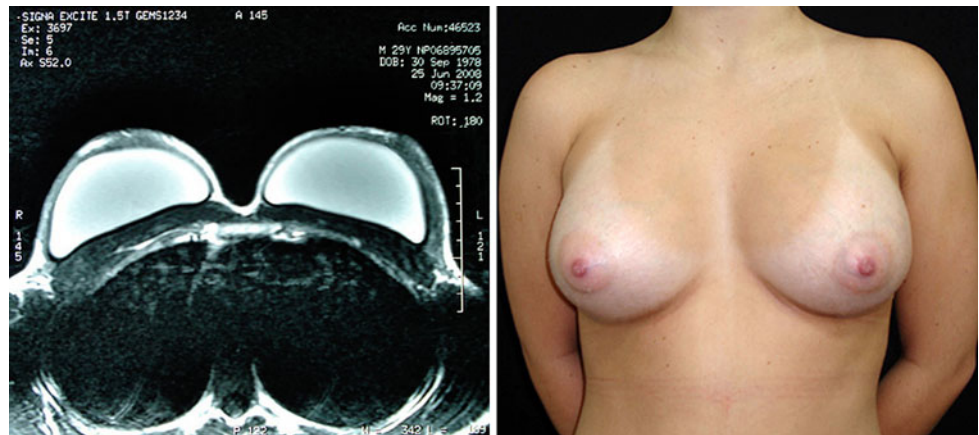


Fig. 2 *Right* Baker grade II in the left breast and Baker grade III in the right with Siltex™ implants after 18 months. *Left* Magnetic resonance imaging confirms the clinical evaluation by the Baker grade

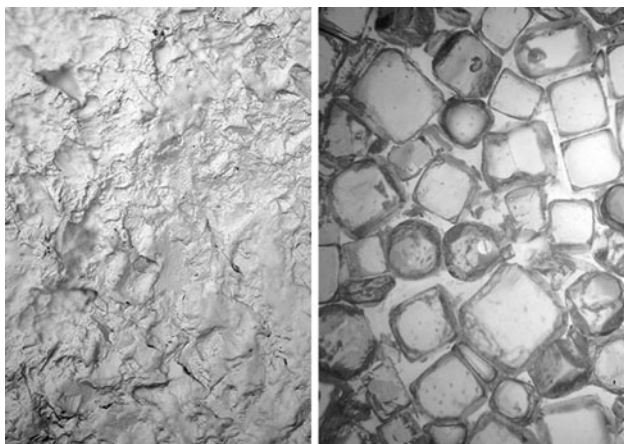
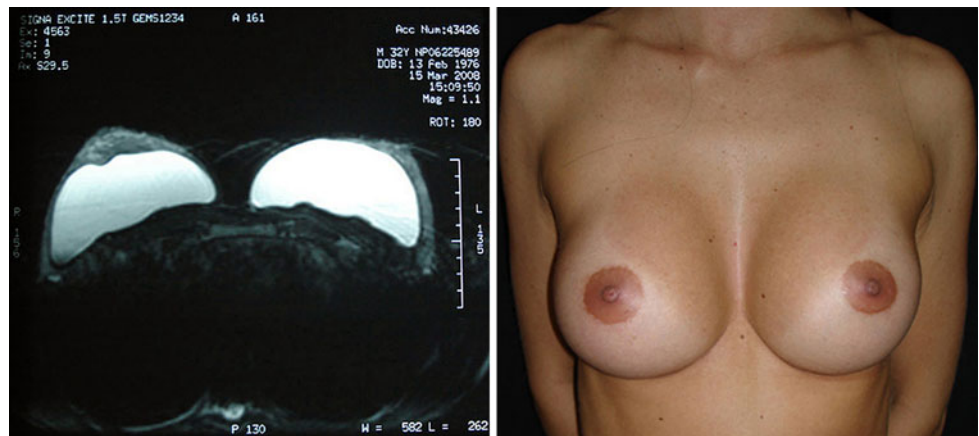


Fig. 3 *Right* Undulations regularly arranged on the microtexture of the Siltex™ implant. *Left* Open-pore diameters from 600 to 800 μm are responsible for the macrotexture of the Biocell™ implant

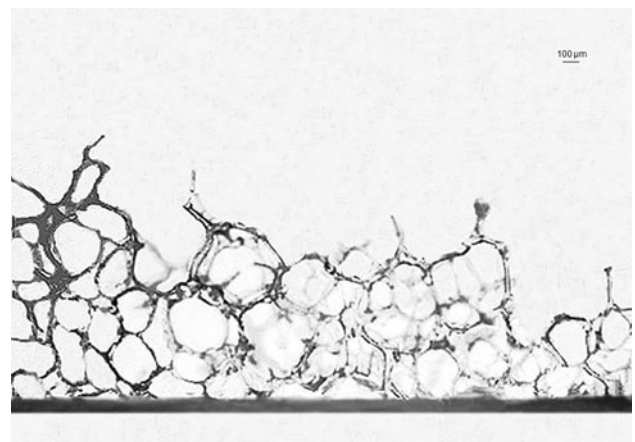


Fig. 4 Biodegradation of the polyurethane foam modified the medium texture of the Polyurethane™ implant creating enlarged open-pore diameters and depths

foam in preventing capsular contracture decreases as it degrades over a period of time, estimated to be 2 years [11]. Biodegradation of the polyurethane broke the regular pattern of the surface texture creating new resultant vectors with different and major lengths and several directions. The increase in length of the resultant vectors induced fibrous

capsular contraction as they came closer to the center of the implant. The undulations of the Siltex™ implant's texture were the origin of very long resultant vectors over the fibrous capsule, increasing breast firmness starting at 9 months. Poepl et al. [12] also achieved early breast firmness for implants with microtextured surfaces. They

Table 2 Volume and texture of the implant

	Volume (ml)	Roughness				
		Surface	Distribution	Density (mm ²)	Open-pore diameter (μm)	Depth (μm)
Siltex™	300	Micronodules	Regular	10	70–150	40–100
Polyurethane™	305	Moderate concavities	Irregular	7	250–350	10–760
Biocell™	300	Macroconcavities	Irregular	5	600–800	150–200

Fig. 5 Parallelogram law applied to linear vectors of macro- and medium-textured implants. *Right* Resultant vectors for Biocell™ showed similar and small lengths with divergent directions. *Left* Resultant vectors for Polyurethane™ had long and different lengths with variable directions

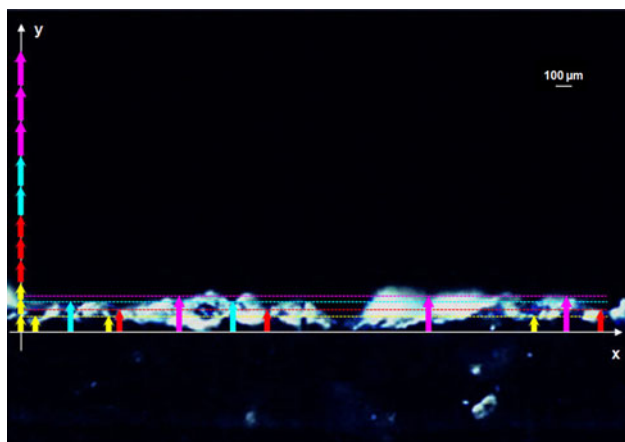
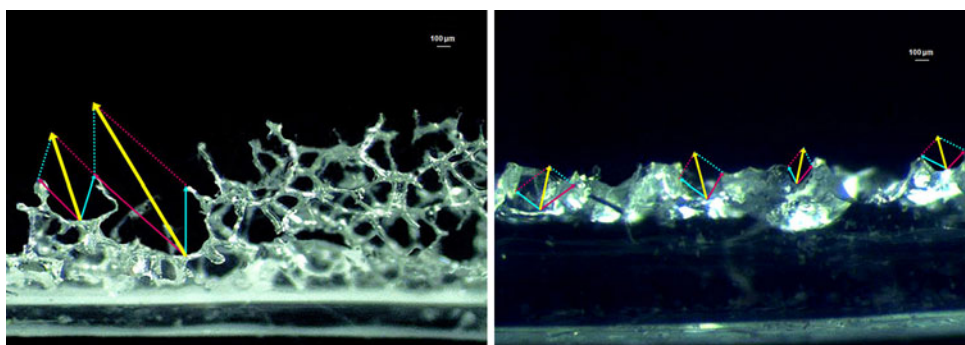


Fig. 6 Siltex™ implants originated linear vectors perpendicular to the horizontal axis of a Cartesian coordinate system. The sum of linear vectors is transposed to the vertical axis creating a single resultant vector perpendicular to the implant surface

found no statistically significant difference between the smooth surface and microtexture surface of the Siltex™ implant.

Unlike Zahavi et al. [13], the histological findings achieved in this study reveal that the thickness of the

fibrous capsule was not a determinative factor for capsular contracture. Despite the fact that Biocell™ implants had more fibrous tissue in the fibrous capsule than did the Polyurethane-Covered Foam™ and Siltex™ implants, the Biocell™ implants showed less capsular contracture. This agrees with the results of Danino et al. [14] who reported the influence of macrotextured surfaces in inducing adhesive effects and resulting in less capsular contraction.

Imaging techniques have been a useful addition to the subjective clinical evaluation of breast firmness, increasing the accuracy of the diagnosis [15]. The MRI done 18 months after breast augmentation supported the subjective evaluation of breast firmness according to the Baker grade. To avoid interference of additional contracture risk factors in capsule formation, we used cohesive silicone gel-filled, 300- and 305-ml implants placed through the inframammary incision and in subglandular position [16].

Conclusion

Despite the small size of our sample, the findings related to vector lengths and direction at the implant texture

Table 3 Fibrous capsule findings

	Macroscopic view	Histological examination		
		Fibrous tissue	Cells	Vessels
Siltex™	Thin linear parallel folds	Linearly organized, slightly thicker	Absent	Absent
Polyurethane™	Irregular distribution, moderate folds	Irregularly disorganized, slightly thicker	Poor	Poor
Biocell™	Adhesive effect	Less compact	High	High

Fig. 7 *Left* Macrotexture of the Biocell™ implant. *Right* Mirror image of the Biocell™ texture imprinted on the interface of the fibrous capsule

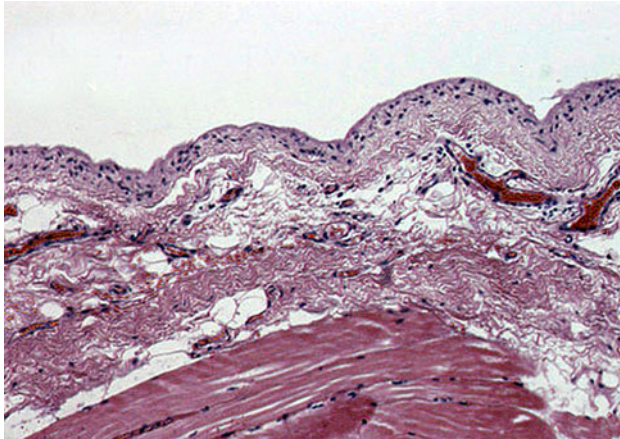
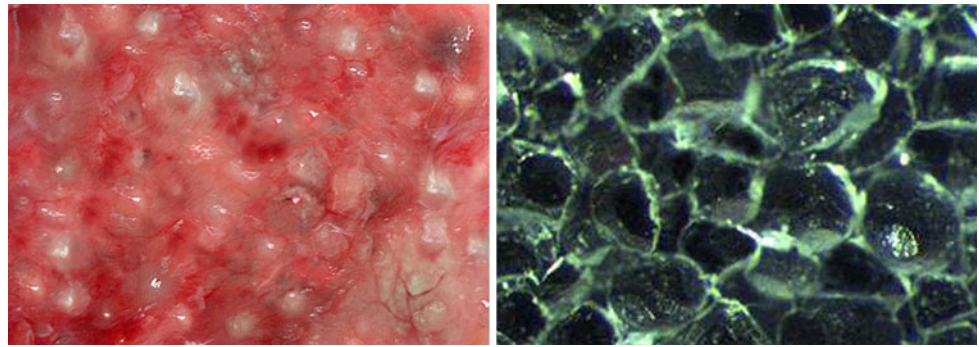


Fig. 8 Mirror image of the Biocell™ texture imprinted over the fibrous capsule. Numerous vessels are shown along the fibrous tissue of the capsule interface

producing a mirror image on the interface of the fibrous capsule, the implant texture, the contour of the capsule around the implant shown on the magnetic resonance imaging, and the Baker grades for capsular contracture were clear. The higher rate of correlation among these findings ensures that the macrotexture of the implant surface significantly reduced capsular contracture.

Acknowledgment Financial support for this work was provided by the ACA—Institute of Assistance in Plastic Surgery of São Paulo.

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